Status of this sheet

The substitution $t=\tan\frac{A}{2}$ (Weierstrass) is **standard RMO/INMO technique** and a bridge to integral calculus; every model below is derivable from Part I identities. Nothing here "can never appear in JEE" — historically JEE-style papers have used it in disguised form; the classification is only that it exceeds the syllabus sentence "Trigonometric functions, general solution, identities".

The substitution and what it buys

Put $t=\tan\dfrac{A}{2}$. Then (for $A\neq\pi+2k\pi$):

$$\sin A=\frac{2t}{1+t^2},\qquad \cos A=\frac{1-t^2}{1+t^2},\qquad \tan A=\frac{2t}{1-t^2}.$$

SHORT DERIVATION

Derivation — divide the double-angle forms by $\cos^2\frac{A}{2}$

$\sin A=2\sin\frac{A}{2}\cos\frac{A}{2}=\frac{2\tan\frac{A}{2}}{\sec^2\frac{A}{2}}=\frac{2t}{1+t^2}$ (using $\sec^2=1+\tan^2$). Similarly $\cos A=\cos^2\frac{A}{2}-\sin^2\frac{A}{2}=\frac{1-t^2}{1+t^2}$ after dividing by $\cos^2\frac{A}{2}$. Tangent: ratio of the two ✓. The point: **any rational expression in $\sin A,\cos A$ becomes a rational function of t** — trig equations become polynomial equations; trig integrals $\int R(\sin x,\cos x)\,dx$ become rational integrals (calculus toolkit).

WORKED EXAMPLE

Worked. Solve $\sin x+\cos x=1$ for $0\le x<2\pi$ with $t=\tan\frac{x}{2}$

$\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}=1\Rightarrow 2t+1-t^2=1+t^2\Rightarrow 2t^2-2t=0$; $t=0$ or $t=1$. $t=0\Rightarrow x=0$; $t=1\Rightarrow\frac{x}{2}=\frac{\pi}{4}\Rightarrow x=\frac{\pi}{2}$. The excluded point $x=\pi$ (where $\cos\frac{x}{2}=0$) gives $\sin\pi+\cos\pi=-1\neq 1$ ✓ nothing lost. Answer: $x\in\{0,\frac{\pi}{2}\}$ ✓.

COMMON TRAP

The blind spot — $A=\pi$ (mod 2π)

$t=\tan\frac{A}{2}$ is undefined when $\frac{A}{2}=\frac{\pi}{2}+k\pi$, i.e. $A=\pi+2k\pi$. Every solution there is invisible to the substitution — **always test $A=\pi+2k\pi$ in the original equation** and add it if it works. For $A\in(-\pi,\pi)$ the map $A\mapsto t$ is a bijection onto $\mathbb{R}$ (this is the clean domain window for the substitution).

KEY RESULT

When to use it. (1) Equations rational in $\sin,\cos$ that resist factorisation; (2) proving identities by clearing denominators in t; (3) converting geometry (slope $=\tan\frac{\theta}{2}$) into algebra; (4) (forward reference) any integral $\int R(\sin x,\cos x)dx$. Counter-indication: when half-angle or product-to-sum is quicker — do the cheap tool first.

Worked models

WORKED EXAMPLE

Worked. Solve $\sin x+\cos x=1$ over all reals

General solutions from the two families: $x=2n\pi$ (from $t=0$) and $x=\frac{\pi}{2}+2n\pi$ (from $t=1$), $n\in\mathbb{Z}$; the blind-spot test $x=\pi+2n\pi$ fails. Combined: $x=2n\pi$ or $x=\frac{\pi}{2}+2n\pi$ ✓ (leave as the two families).

WORKED EXAMPLE

Worked. Express $\frac{1-\cos x}{\sin x}$ in t

$\frac{1-\cos x}{\sin x}=\frac{1-\frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}}=\frac{\frac{2t^2}{1+t^2}}{\frac{2t}{1+t^2}}=t=\tan\frac{x}{2}$ ✓ — consistent with the half-angle identity of Part I. Any rational $R(\sin x,\cos x)$ now reduces by t-clearing denominators; $1+t^2\neq 0$ is automatic for real t.

Assumptions & conventions

A real with $A \neq \pi + 2k\pi$ under the substitution; the blind spot is checked separately in every problem; rational expressions in $\sin, \cos$ mean ratios of polynomials; all conversions verified back in the original equation.

Status of this sheet

*Products like $\prod \sin \frac{k\pi}{n}$ are **classic RMO/INMO algebra** obtained by combining Part I identities with complex numbers (chapter 5 of the main sequence supplies $z = re^{i\theta}$ and $|z|$; only elementary modulus facts are used).*
Classification: beyond the syllabus sentence, but every step is derivable from supplied identities and the definition of $|1 - e^{i\theta}|$.

The two flagship products

$$\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$$

$$\prod_{k=1}^{n} \cos \frac{k\pi}{2n+1} = \frac{1}{2^n}$$

SHORT DERIVATION

Derivation — factor $z^n - 1$ at the roots of unity, set $z = 1$, take moduli

The n roots of $z^n = 1$ are $e^{2\pi ik/n}$, $k = 0, \ldots, n-1$. So $z^n - 1 = (z-1)\prod_{k=1}^{n-1}(z - e^{2\pi ik/n})$. Cancel $z - 1$ ($\frac{z^n-1}{z-1} = 1 + z + \cdots + z^{n-1}$) and set $z = 1$: $n = \prod_{k=1}^{n-1}(1 - e^{2\pi ik/n})$. Now $|1 - e^{i\theta}| = 2\left|\sin \frac{\theta}{2}\right|$ (chord of a unit-circle arc; expand $|1 - \cos\theta - i\sin\theta|$). Taking moduli: $n = \prod 2\sin \frac{k\pi}{n} = 2^{n-1}\prod \sin \frac{k\pi}{n}$ (each $\sin \frac{k\pi}{n} > 0$ for $1 \le k \le n-1$) ✓. The cosine product comes from the same factorisation at $z = -1$.

WORKED EXAMPLE

Worked. $\prod_{k=1}^{6} \sin \frac{k\pi}{7}$ and $\cos \frac{\pi}{7}\cos \frac{2\pi}{7}\cos \frac{3\pi}{7}$

$n = 7$: product $= \frac{7}{2^6} = \frac{7}{64} = 0.109375$ ✓ (direct: $0.43388 \times 0.78183 \times 0.97493 \times 0.97493 \times$). Cosine product with $2n + 1 = 7$ ($n = 3$): $\frac{1}{2^3} = \frac{1}{8}$ ✓ (direct: $0.90097 \times 0.62349 \times 0.22252 = 0.125$).

The pattern to internalise

A product of trig values at rational multiples of π is a roots-of-unity problem in disguise.
*Second tool: when the angles come from a polynomial whose roots are $\tan$-values or the values themselves, **Vieta** hands you sums/products for free (see the $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3}$ model of Part I sheet 8 — same idea with $\tan 3\theta$).*

WORKED EXAMPLE

Worked. Sine product for $n = 5$: $\sin 36^\circ \sin 72^\circ$?

$\prod_{k=1}^{4} \sin \frac{k\pi}{5} = \frac{5}{16}$. Pair $\sin \frac{\pi}{5} = \sin 36^\circ$, $\sin \frac{2\pi}{5} = \sin 72^\circ$, $\sin \frac{3\pi}{5} = \sin 72^\circ$, $\sin \frac{4\pi}{5} = \sin 36^\circ$: product $= (\sin 36^\circ \sin 72^\circ)^2 = \frac{5}{16}$, so $\sin 36^\circ \sin 72^\circ = \frac{\sqrt{5}}{4} \approx 0.5590$ ✓ (direct: $0.58779 \times 0.95106 = 0.55902$).

COMMON TRAP

Traps on this sheet

- $\prod_{k=1}^{n-1}$ vs $\prod_{k=1}^{n}$: the sine product has $n-1$ factors (the $k=0$ factor would be $\sin 0 = 0$); count factors before quoting.
- $|1-e^{i\theta}| = 2|\sin\frac{\theta}{2}|$ — drop the absolute values only after checking positivity (done via $0 < \frac{k\pi}{n} < \pi$).
- The cosine product $\prod_{k=1}^{n}\cos\frac{k\pi}{2n+1} = 2^{-n}$ has n factors and denominator $2n+1$ — a different index bookkeeping from the sine product.
- Claims here need complex arithmetic: state $e^{i\theta} = \cos\theta + i\sin\theta$ and $|z|$ facts explicitly in a solution write-up (assumptions below).

Assumptions & conventions

Complex numbers used: $e^{i\theta} = \cos\theta + i\sin\theta$, $|zw| = |z||w|$, $|1-e^{i\theta}| = 2|\sin\frac{\theta}{2}|$ (proved above in modulus form). All products finite; $n \geq 2$ integer; positivity of $\sin\frac{k\pi}{n}$ for $1 \leq k \leq n-1$ used to drop absolute values.

Status of this sheet

The telescoping series $\sum\sin k\theta$ and the sums $\sum\cot^2$, $\sum\csc^2$ are ***standard olympiad algebra***; all derive from Part I transformation formulae and the roots-of-Vieta technique of sheet 2. Models are derivable from supplied information; classified as extensions because they sit beyond the current syllabus sentence.

Sums with arguments in arithmetic progression (telescoping)

The trick: multiply by $2\sin\frac{\theta}{2}$ and watch the product-to-sum formula telescope.

$$\sum_{k=1}^{n}\sin k\theta = \frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

$$\sum_{k=1}^{n}\cos k\theta = \frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

SHORT DERIVATION

Derivation — the $2\sin\frac{\theta}{2}$ multiplier

$2\sin\frac{\theta}{2}\sin k\theta = \cos\left(k-\frac{1}{2}\right)\theta - \cos\left(k+\frac{1}{2}\right)\theta$. Summing over $k = 1\ldots n$ telescopes to $\cos\frac{\theta}{2} - \cos\left(n+\frac{1}{2}\right)\theta$. Then apply $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$ with $C = \frac{\theta}{2}$, $D = (n+\frac{1}{2})\theta$, $= 2\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}$. Divide by $2\sin\frac{\theta}{2}$ ✓. (Valid for $\sin\frac{\theta}{2} \neq 0$, i.e. $\theta \notin 2\pi\mathbb{Z}$; the excluded cases are computed directly as sums of 0/1.)

WORKED EXAMPLE

Worked check. $n = 3$, $\theta = \frac{\pi}{6}$

Direct: $\sin 30^\circ + \sin 60^\circ + \sin 90^\circ = 0.5 + 0.86603$. Formula: $\frac{\sin 45^\circ \sin 60^\circ}{\sin 15^\circ} = \frac{0.70711 \times 0.86603}{0.25882} = 2.36603$ ✓.

Sums of squares at rational multiples of π (roots + Vieta)

The numbers $t_k = \tan\frac{k\pi}{n}$, $k = 1, \ldots, n-1$, are the non-zero roots of a polynomial obtained from $\tan(n\theta)$ in terms of $t = \tan\theta$ (write $\tan n\theta = \frac{\Im(1+it)^n}{\Re(1+it)^n}$ and set the numerator to 0). Vieta then gives sums of powers.

$$\sum_{k=1}^{n-1}\cot^2\frac{k\pi}{n} = \frac{(n-1)(n-2)}{3}, \qquad \sum_{k=1}^{n-1}\csc^2\frac{k\pi}{n} = \frac{n^2-1}{3}.$$

SHORT DERIVATION

Derivation sketch — cotangents as roots of the complementary polynomial

$\cot^2\frac{k\pi}{n}=\frac{1}{t_k^2}$, and the symmetric identity $\sum\frac{1}{t_k^2}=\left(\sum\frac{1}{t_k}\right)^2-2\sum_{j<k}\frac{1}{t_jt_k}$ turns Vieta's data for the t_k into the closed form $\frac{(n-1)(n-2)}{3}$. Then $\csc^2=1+\cot^2$ adds $(n-1)$ terms of 1: $\sum\csc^2=(n-1)+\frac{(n-1)(n-2)}{3}=\frac{(n-1)(n+1)}{3}=$ ✓.

WORKED EXAMPLE

Worked checks. $n=3,4,5$ $\cot^2$: $n=3$: $\cot^2 60^\circ+\cot^2 120^\circ=\frac{1}{3}+\frac{1}{3}=\frac{2}{3}=\frac{2\cdot1}{3}$ ✓. $n=4$: $1+0+1=2=\frac{3\cdot2}{3}$ ✓. $n=5$: $4=\frac{4\cdot3}{3}$ ✓. $\csc^2$: $n=3$: $\frac{4}{3}+\frac{4}{3}=\frac{8}{3}=\frac{9-1}{3}$ ✓; $n=4$: $2+1+2=5=\frac{15}{3}$ ✓; $n=5$: $8=\frac{24}{3}$ ✓.

KEY RESULT

Whenever a sum runs over trig values at equally spaced angles: (1) look for a telescoping multiplier ($2\sin\frac{\theta}{2}$ for AP-argument sums); (2) look for the polynomial whose roots they are (Vieta for square/power sums); (3) pair symmetric terms ($k\leftrightarrow n-k$) to halve the work; (4) validate a small case numerically before the proof.

COMMON TRAP

Traps on this sheet

- The telescoping formulas divide by $\sin\frac{\theta}{2}$; state $\theta\notin 2\pi\mathbb{Z}$ (or handle those cases directly).
- $\sum\sin k\theta$ closed form: the numerator is $\sin\frac{n\theta}{2}\cdot\sin\frac{(n+1)\theta}{2}$ — order $(n+1)$ vs n swapped between the two factors vs the cosine version (which has $\cos\frac{(n+1)\theta}{2}$).
- $\cot^2\frac{k\pi}{n}$ has a term at $k=\frac{n}{2}$ (when n even) equal to 0 — include it (it is $\cot^2\frac{\pi}{2}=0$).
- $\tan n\theta$'s polynomial in $t=\tan\theta$ has **degree** n in the tangent-angle formula bookkeeping; only the roots t_k with $\tan$ defined count.

Assumptions & conventions

$n\geq 2$ integer; θ real with $\sin\frac{\theta}{2}\neq 0$ in series formulas; Vieta applied to real polynomials; numeric checks above computed in radians (converted to degrees for readability).

Status of this sheet

Important classification note. The current JEE Advanced syllabus sentence for trigonometry is "Trigonometric functions. General solution of trigonometric equations. Identities." — it does **not** list properties of triangles or heights & distances. The formulae below are the **standard triangle toolkit** used across JEE Main-level papers, olympiads (RMO/INMO geometry-algebra), and later chapters (conics, 3D, vectors). They are added theory (not derivable from the identities alone — they encode triangle geometry), but every model built here only uses Part I identities plus the sine/cosine rules as stated. Do not assume they "can never appear"; assume they are simply outside the quoted syllabus sentence and learn them as toolkit.

The four triangle formulae

For a triangle with sides a, b, c opposite angles A, B, C, circumradius R, inradius r, semiperimeter $s = \frac{a+b+c}{2}$, area Δ:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \qquad a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}bc\sin A = \sqrt{s(s-a)(s-b)(s-c)}, \qquad r = \frac{\Delta}{s}$$

$$\tan\frac{A}{2} = \frac{r}{s-a}$$

SHORT DERIVATION

Cosine rule and $\Delta = rs$ — one line each

Cosine rule: drop a height $h = c\sin B$ from C; project: $a = b\cos C + c\cos B$-style coordinate computation gives $a^2 = b^2 + c^2 - 2bc\cos A$ (or expand $|\vec{B} - \vec{C}|^2$ in the vectors chapter). $\Delta = rs$: split the triangle into three (r, s)-shaped pieces — $\Delta = \frac{1}{2}ra + \frac{1}{2}rb + \frac{1}{2}rc = rs$ ✓. $\tan\frac{A}{2} = \frac{r}{s-a}$: the incircle tangent lengths are $s-a, s-b, s-c$ and half-angle right triangles give the ratio.

Derived special values (pure identities, pentagon angles)

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4} \approx 0.30902$$

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4} \approx 0.80902$$

SHORT DERIVATION

Derivation — double-angle, then quadratic

Let $x = 18^\circ$, so $5x = 90^\circ$ and $2x = 90^\circ - 3x$, hence $\sin 2x = \cos 3x$. Expand: $2\sin x\cos x = 4\cos^3 x - 3\cos x$; with $\cos x \neq 0$: $2\sin x = 4\cos^2 x - 3 = 4(1 - \sin^2 x) - 3 = 1$. So $4\sin^2 x + 2\sin x - 1 = 0 \Rightarrow \sin x = \frac{-2+\sqrt{4+16}}{8}$ (positive root; $\sin 18^\circ > 0$) ✓. Then $\cos 36^\circ = 1 - 2\sin^2 18^\circ = \frac{\sqrt{5}+1}{4}$ ✓ (check numerically: 0.809017).

WORKED EXAMPLE

Worked. Triangle with $A = 60^\circ$, $b = 5$, $c = 8$; find a and Δ

Cosine rule: $a^2 = 25 + 64 - 2\cdot 5\cdot 8\cdot\frac{1}{2} = 49 \Rightarrow a = 7$.

$\Delta = \frac{1}{2}\cdot 5\cdot 8\cdot\frac{\sqrt{3}}{2} = 10\sqrt{3} \approx 17.32$.

Cross-check with Heron: $s = \frac{7+5+8}{2} = 10$, $\sqrt{10\cdot 3\cdot 5\cdot 2} = \sqrt{300} = 10\sqrt{3}$ ✓.

WORKED EXAMPLE

Worked. R and r for the same triangle

$2R = \frac{a}{\sin A} = \frac{7}{\sqrt{3}/2} = \frac{14}{\sqrt{3}} \Rightarrow R = \frac{7}{\sqrt{3}} \approx 4.041$.

$r = \frac{\Delta}{s} = \frac{10\sqrt{3}}{10} = \sqrt{3} \approx 1.732$. Check $\tan\frac{A}{2} = \tan 30^\circ = \frac{1}{\sqrt{3}}$ vs $\frac{r}{s-a} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$ ✓.

COMMON TRAP

Traps — triangle toolkit

- The sine rule gives an **ambiguous case** (SSA): $a = b\frac{\sin A}{\sin B}$ with $\sin B = \frac{b\sin A}{a}$ may admit two triangles (B and $180^\circ - B$).
- $\Delta = \frac{1}{2}(\text{base})(\text{height})$ needs the height; $\frac{1}{2}bc\sin A$ needs the angle **between** the two sides.
- Heron's formula: s is semiperimeter; writing $s = a + b + c$ is the classic slip.
- $\tan\frac{A}{2} = \frac{r}{s-a}$ requires the usual triangle convention ($s > a$); for degenerate/failing configurations state the inequality.
- 18°-family values ($\sin 18^\circ, \cos 36^\circ, \sin 54^\circ = \cos 36^\circ$ etc.): derive from the quadratic each time if memory fails — and pick the sign from the quadrant (18° is QI).

Assumptions & conventions

Triangle = non-degenerate (positive angles summing to π, positive sides satisfying triangle inequalities); R circumradius, r inradius, s semiperimeter, Δ area; special values derived from double-angle identities as shown; toolkit status per the note at the top of this sheet.